```
#include <iostream>
#include <queue>
using namespace std;

struct Node
{
   int data;
   Node *left, *right;
};

Node* createNode(int value)
{
   Node* newNode = new Node;
   newNode->data = value;
   newNode->left = NULL;
   newNode->right = NULL;
   return newNode;
}

int height(Node* root, int &dia)
{
   if (root == NULL)
      return 0;

   int leftH = height(root->left, dia);
   int rightH = height(root->right, dia);

   dia = max(dia, leftH + rightH);

   return 1 + max(leftH, rightH);
}
```

```
int maxWidth(Node* root)
{
    if (root == NULL)
        return 0;

    queue<Node*> q;
    q.push(root);
    int maxW = 0;

    while (!q.empty())
    {
        int count = q.size();
        maxW = max(maxW, count);

        for (int i = 0; i < count; i++)
        {
            Node* temp = q.front();
            q.pop();

            if (temp->left != NULL)
                q.push(temp->left);

            if (temp->right != NULL)
                q.push(temp->right);
        }
    }

    return maxW;
}
```

```
int main()
{
    Node* root = createNode(1);
    root->left = createNode(2);
    root->right = createNode(3);
    root->left->left = createNode(4);
    root->left->right = createNode(5);
    root->right->left = createNode(6);

    int dia = 0;
    height(root, dia);

    cout << "Computer Network Analysis" << endl;
    cout << "Diameter: " << dia << " edges" << endl;
    cout << "Maximum Width: "
        << maxWidth(root) << " nodes" << endl;

    return 0;
}
```